# SIMATS ENGINEERING

## ASSESSMENT TOOL 3: Technical Presentation

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

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**COURSE OUTCOME COVERED**

**CO1:** *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**
Question Count: 5

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**Code of Conduct**

I Sanjana.G (192511020) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Sanjana.G*

**Assessment Tasks**

**Technical Presentation Topics**

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

Here are 10 additional technical presentation topics in a similar style, suitable for a 6–8 minute presentation with definitions, architecture, industry relevance, and a practical example:

1. **Containerization and Cloud-Native Applications: Transforming Modern Software Deployment**
   - Explore containers, container architecture, benefits, and a real-world deployment example.
2. **Serverless Computing: Reducing Infrastructure Management in the Cloud**
   - Discuss Function-as-a-Service (FaaS), architecture, business advantages, and practical use cases.
3. **Edge Computing vs Cloud Computing: Choosing the Right Computing Paradigm**
   - Compare architectures, applications, and industry adoption scenarios.
4. **Microservices Architecture in Cloud Environments**
   - Explain microservices design, benefits over monolithic systems, and industry implementations.
5. **Cloud Security Fundamentals: Protecting Data and Applications**
   - Cover security models, shared responsibility architecture, and real-world security practices.
6. **Multi-Cloud and Hybrid Cloud Strategies for Enterprise IT**
   - Analyze deployment models, business benefits, challenges, and practical examples.
7. **The Role of Artificial Intelligence and Machine Learning in Cloud Platforms**
   - Discuss AI/ML cloud services, architecture, and industry applications.

8. **DevOps and Continuous Integration/Continuous Deployment (CI/CD) in Cloud Computing**
    - Explain DevOps principles, CI/CD architecture, and practical implementation examples.
9. **Cloud Storage Technologies: Object, Block, and File Storage Compared**
    - Compare storage architectures, use cases, and industry adoption patterns.
10. **Internet of Things (IoT) and Cloud Computing: Building Connected Ecosystems**
    - Examine IoT-cloud architecture, benefits, challenges, and real-world applications.
11. **Cloud Migration Strategies: Moving Enterprise Applications to the Cloud**
    - Examine migration approaches, cloud architecture considerations, benefits, challenges, and a successful migration case study.
12. **Big Data Analytics in Cloud Computing**
    - Explore cloud-based data processing architectures, analytics platforms, business value, and real-world applications.
13. **Identity and Access Management (IAM) in Cloud Environments**
    - Discuss authentication, authorization architectures, security benefits, and practical enterprise implementations.
14. **Content Delivery Networks (CDNs) and Their Role in Cloud Services**
    - Explain CDN architecture, performance optimization, industry relevance, and examples from popular web platforms.
15. **Server Virtualization and Its Role in Cloud Computing**

    Explain virtualization architecture, hypervisors, virtual machines, resource allocation, and the benefits of virtualization in cloud data centers.

16. **Cloud Storage Technologies and Architecture**

    Discuss cloud storage concepts, storage architectures, scalability, availability, data replication, and practical applications such as Google Drive and Amazon S3.

17. **Cloud Security: Threats and Protection Mechanisms**

    Explain common cloud security threats, data protection, encryption, network security, security controls, and practical approaches used by cloud service providers.

18. **Infrastructure as a Service (IaaS): Architecture and Applications**

    Explain the IaaS model, virtual computing resources, storage and networking services, advantages, limitations, and practical enterprise applications.

19. **Platform as a Service (PaaS) for Cloud Application Development**

    Discuss PaaS architecture, development environments, application deployment, scalability, advantages, limitations, and examples of PaaS platforms.

20. **Software as a Service (SaaS): Architecture and Business Applications**

    Explain SaaS architecture, features, advantages, subscription-based software delivery, security considerations, and examples such as Google Workspace and Microsoft 365.

**Minimum slide flow:**

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway

**Technical Presentation Rubric**

| Criteria | Weightage | Level 4 - Exemplary | Level 3 - Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Technical Content | 30% | Content is highly accurate, relevant, and insightful | Content is correct with minor gaps | Basic content with limited depth | Content is inaccurate or superficial |
| Structure and Flow | 20% | Excellent logical flow and slide organization | Good structure with minor issues | Some organization but weak flow | Disorganized presentation |
| Use of Visuals | 15% | Visuals strongly support technical communication | Relevant visuals used well | Limited or unclear visuals | No meaningful visuals |
| Communication Skills | 20% | Confident, clear, and engaging delivery | Clear delivery with minor hesitation | Moderate clarity but uneven delivery | Weak delivery and low clarity |
| Professionalism | 15% | Well-prepared and audience-focused presentation | Mostly professional | Basic preparation visible | Poor preparation and professionalism |

**Topic :**

**Introduction**

**Objective**

**Core technical explanation:**

**Architecture or process diagram:**

**Real-world use case:**

**Conclusion and key takeaway:**

**Photo of the presentation :**

**Conclusion :**

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Engineer to Excel

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**TECHNICAL PRESENTATION**

# The Role of Artificial Intelligence and Machine Learning in Cloud Platforms

## TEAM NO:7

### CSA1520 – Cloud Computing and Big Data Analytics


**Guided By:**

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## Abstract


- AI and ML are becoming important in cloud computing.
- AI can help manage cloud resources automatically.
- ML can analyze large amounts of cloud data.
- AI can help reduce cloud costs.
- It can also improve the speed and performance of cloud services.

# Aim and Objective

**Primary Aim**

To architect and evaluate an intelligent, self-scaling AI/ML cloud platform that optimizes resource allocation and delivers low-latency predictive analytics.

### Objective 1

- Evaluate AutoML frameworks vs manual hyperparameter tuning on cloud clusters.

### Objective 2

- Construct robust MLOps pipelines for automated model deployment and drift tracking.

### Objective 3

- Implement Reinforcement Learning for real-time dynamic GPU auto-scaling.

## Introduction

- Cloud computing provides storage and computing services through the internet.
- Earlier, cloud systems mainly provided basic storage and computing.
- Today, AI and ML are integrated into cloud platforms.
- Major cloud providers include AWS, Microsoft Azure and Google Cloud.
- AI helps cloud systems become smarter and more automatic.

# Core Technical Explanation – Define the Problem

### Identified Bottlenecks

- **Static Provisioning Inefficiency:** Fixed GPU instances leading to 45% idle resource wastage.
- **Data Ingestion Latency:** Unoptimized I/O pipelines bottlenecking high-throughput deep learning.
- **Model Drift Vulnerability:** Undetected concept drift degrading inference accuracy over time.

### Insights from Literature Review

- **Legacy Reactive Auto-Scalers:** Threshold-based triggers fail under rapid traffic spikes.
- **Siloed MLOps Tooling:** Lack of standardized pipelines between data science and ops teams.
- **High Cloud Overhead:** Uncontrolled GPU instance sprawl driving up operational expenditures.

## Existing System

- Cloud resources are often managed manually.
- Scaling resources depends on fixed rules.
- Models may require manual deployment.
- Resources can be overused or underused.
- There is less automatic monitoring and recovery.

## Proposed System

- AI can automatically manage cloud resources.
- Resources can increase when demand is high.
- Resources can decrease when demand is low.
- ML models can be monitored automatically.
- This can reduce cost and improve performance.

# Overall System Architecture Design

**Layer 1: Data Ingestion** — Streaming Ingestion (Apache Kafka) + Cloud Lakehouse (AWS S3 / GCP BigQuery)

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**Layer 2: AI Core & AutoML** — Feature Store + AutoML Optimization + Hyperparameter Tuning Engine

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**Layer 3: Orchestration & Compute** — Kubernetes GPU Pods + RL Dynamic Resource Scheduler (RL-DRMS)

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**Layer 4: Serving & MLOps** — RPC / REST Gateway + Real-time Model Drift Monitor (Prometheus)

# Proposed Methodology and Algorithm

**Algorithm Name:** **RL-DRMS (Reinforcement Learning-based Dynamic Resource & Model Scheduling)**

### Step 1

**Telemetry Extraction**

- Monitor GPU utilization, memory bandwidth, query queue length.

### Step 2

**Reward Calculation**

- Reward = SLA Compliance - Cloud Billing Expense.

### Step 3

**Action Execution**

- Dynamically spin up spot GPU instances or prune active workers.

### Step 4

**AutoML Retraining**

- Trigger model retraining if concept drift exceeds 5% error.

# Real-World Use Case in Cloud Computing

**Healthcare example:**

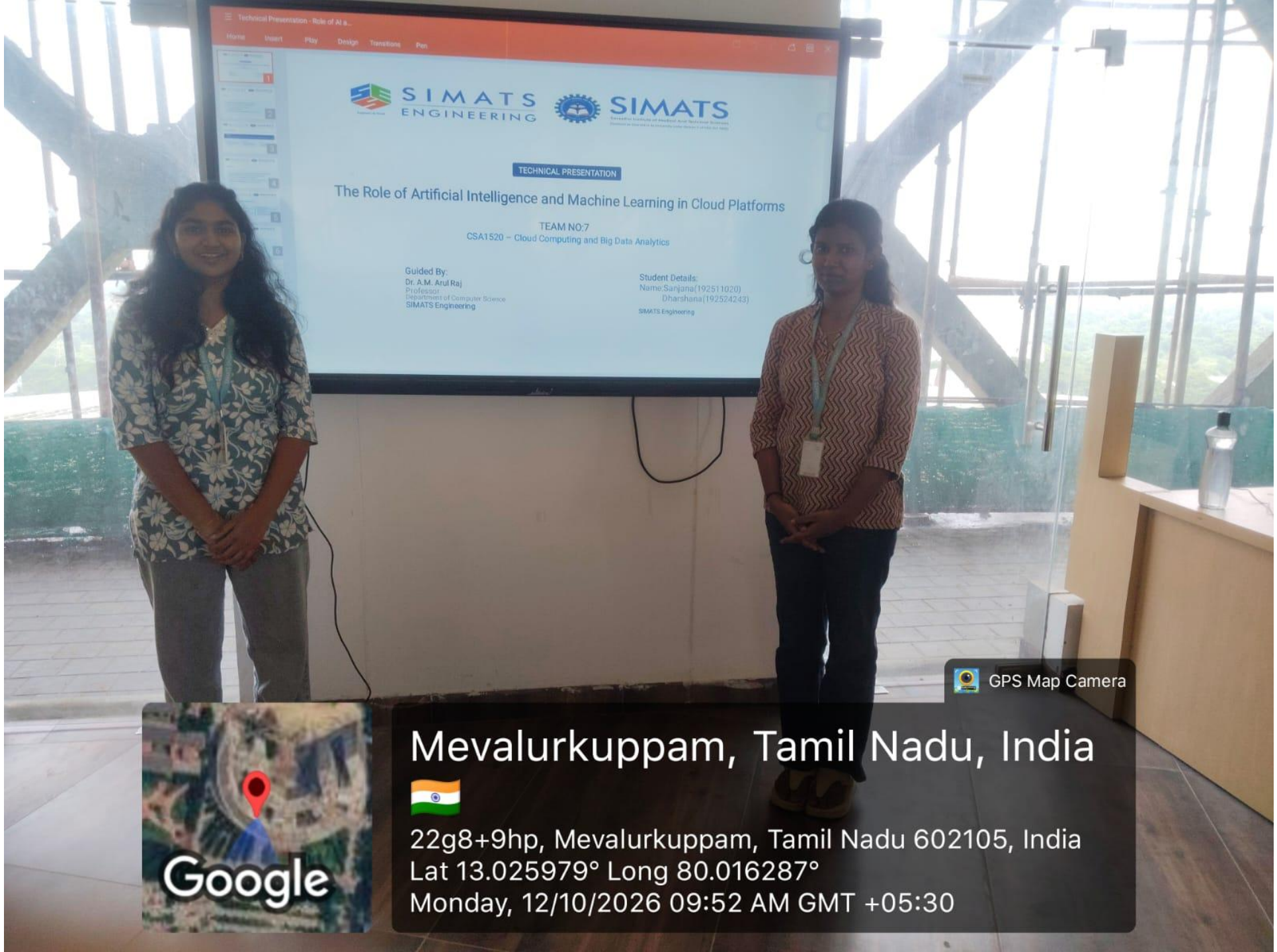
- Medical images can be stored in the cloud.
- AI can analyze these images.
- ML can help identify possible diseases.
- Cloud resources can automatically increase when many images need to be processed.
- This can save time and improve efficiency.

## Conclusion

- AI and ML make cloud platforms smarter.
- AI can automatically manage cloud resources.
- ML can analyze data and improve predictions.
- Automation can reduce cost and save time.
- AI-based cloud computing can be useful in many fields.

# References


- Dean, J., et al. (2023). "Machine Learning Infrastructure at Hyperscale." *IEEE Micro*, 43(2), pp. 14-25.
- Zhang, Y., & Kumar, R. (2024). "Reinforcement Learning for Dynamic Cloud Resource Allocation." *ACM Computing Surveys*, 56(4), pp. 1-38.
- Amazon Web Services. (2025). "Architecting MLOps Workloads on AWS SageMaker." AWS Enterprise Whitepaper.
- Google Cloud Platform. (2024). "Vertex AI Architecture and Best Practices for Enterprise Scaling." Google Cloud Architecture Center.
- Vaswani, A., et al. (2023). "Distributed AI Systems in Multi-Cloud Environments." *IEEE Transactions on Cloud Computing*, 11(3), pp. 2105-2118.


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